

Production functional realisation – standalone backend closure

TECT verification-first repository · claim A1-PRODUCTION-FUNCTIONAL-REALISATION

first issued 2026-07-17 · this version issued 2026-07-17 · v1.2

1. Purpose and scope

The manifest-defined three-component complex functional is now implemented in a standalone, hash-pinned backend. A separate verifier reproduces the discrete variational identities on spectral torus grids $N = 4, 6, 8$. This closes the P1 technical implementation-and-verification task at T4 strong evidence. It does not repair the separately pinned historical working branch, and it does not assert a continuum theorem, minimizer, BCC selection, or stability result.

2. Frozen functional

With $\rho = \Psi^\dagger \Psi$, $m_A = \Psi^\dagger T_A \Psi$, $J_A = \nabla m_A$, and $K_A = J_A - (m_A/(\rho + 10^{-12}))\nabla \rho$, the standalone backend computes

$$F = F_{\text{Braz}} + F_{\text{family}} + F_{\text{lock}} + F_{\text{shell}} + \int \left(\frac{\lambda}{4} \rho^2 + \frac{\gamma}{6} \rho^3 \right) + \sum_A \int \left(\frac{a}{2} |J_A|^2 + b \operatorname{Re}(\bar{J}_A \cdot K_A) + \frac{c}{2} |K_A|^2 \right),$$

where $a = c_{JJ} \alpha_X^2 / (M_X^2 + 10^{-12})$, $b = c_{JK} \alpha_X \beta_X / (M_X^2 + 10^{-12})$, and $c = c_{KK} \beta_X^2 / (M_X^2 + 10^{-12})$. The implementation includes the Brazovskii linear term, family masses, internal lock, shell bias, quartic and sextic terms, all three Class-II coefficients, the density regularizer, and the manifest projector conventions.

3. Independent verification matrix

The verifier imports only the standalone backend and independently defines the real torus pairing, field generators, finite differences, and analytic one-component scalar residual. It tests zero, homogeneous, random, q_0 -shell, and Class-II-active fields at the pinned parameter point and at the declared shell-bias activation $\eta_{\text{shell}} = 0.07$. Each grid uses finite difference steps $10^{-4}, 10^{-5}, 10^{-6}$.

All gates pass. Across $N = 4, 6, 8$, the maximum relative errors are

$$\max |DF - \langle R, v \rangle_{\mathbb{R}}|_{\text{rel}} = 1.856661591692621 \times 10^{-7},$$

$$\max \|DRv - Hv\|_{\text{rel}} = 6.695217715947473 \times 10^{-9}.$$

$$\max |\langle u, Hv \rangle_{\mathbb{R}} - \langle Hu, v \rangle_{\mathbb{R}}|_{\text{rel}} = 1.282693395943113 \times 10^{-10}.$$

The exact scalar-reduction maximum is $6.218564371786818 \times 10^{-16}$. The persisted verifier assertion list also checks source existence, source hash, source version, grid count, required field coverage, and nonzero activation of every listed coupling.

4. Reproduction record

The direct multi-grid run is stored at [runs/2026-07-17-production-backend-multigrid/result.json](#). The clean subprocess preflight at [runs/promotion-evidence/2026-07-17-local-preflight-v3/](#) records the Python environment, git state, input hashes, stdout, stderr, full verifier JSON, review checklist, and file hashes. Its verdict is `TECHNICAL-PASS`; by construction, local preflight is not independent reproduction evidence.

5. Devil’s-advocate

(alpha) “*The Class-II cross coupling is omitted or conflated with the K square.*” `DISMISSED`: c_{JJ} , c_{JK} , and c_{KK} enter three distinct quadratic coefficients and all are nonzero in the activation check.

(beta) “*Autodiff makes every check tautological.*” `VALID` with mitigation: the verifier independently computes energy finite differences, outer residual differences, the real pairing, and the analytic scalar reduction. Torch supplies the backend residual, not the verification formulas.

(gamma) “*Three small grids prove the continuum functional.*” `UPHELD` as an invalid reading: the certified object is the stated discrete spectral implementation only. Grid convergence and continuum well-posedness are absent.

(delta) “*This silently validates the historical production solver.*” `UPHELD` as an invalid reading: the historical external source remains frozen as `EXTERNAL-SOURCE-AUDIT-FAIL`. Solver integration is a separate action.

(epsilon) “*The local preflight is an independent T5 reproduction.*” `UPHELD` as an invalid reading: the evidence collector marks it non-certifying. An operator-named independent run and final confirmed bundle are still required for any T5 decision.

6. Result footer

Result ID:	A1-PRODUCTION-FUNCTIONAL-REALISATION
Precise statement:	Hash-pinned standalone all-coupling backend passes the discrete variational and scalar-reduction gates.
Scope:	Spectral torus grids N=4,6,8; five field classes; pinned parameters plus declared shell activation.
Dependencies:	A1-PRODUCTION-KERNEL-MANIFEST.
Evidence grade:	EXECUTED, INDEPENDENT-CALCULATION, HASH-PINNED.
Direct command:	python codes/foundations/a1_production_backend_verify.py --grids 4 6 8
Expected output:	PRODUCTION-BACKEND-MULTIGRID-PASS.
Preflight command:	python codes/foundations/a1_functional_promotion_evidence.py --run-id <id> --reviewer <name> --mode preflight
Falsification gate:	source/hash drift or any failed identity, field, grid, coupling-activation, or scalar-reduction assertion.
Tier before / after:	T3 / T4.
No-overclaim statement:	No external-source repair, continuum theorem, minimizer, BCC, stability, or T5/T6/T7 result.
Next required action:	independent operator reproduction before scoped T5 review.